MSG No. 130.428 $P4'/n'c'c$ [Type III, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,4,11,12]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,9,10]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[5,6,14,15]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,8,13,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,10,13,16]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[11,12,14,15]`

Table 3: Wyckoff site: `4c`, site symmetry: `4'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,4,9,10]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[2,3,11,12]`
3	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[5,6,13,16]`
4	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[7,8,14,15]`

Table 4: Wyckoff site: `8d`, site symmetry: `-1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,13]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,14]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,15]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,16]`
5	$\left[\frac{1}{2}, 0, 0\right]$	`[5,9]`
6	$\left[0, \frac{1}{2}, 0\right]$	`[6,10]`
7	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,11]`
8	$\left[0, 0, \frac{1}{2}\right]$	`[8,12]`

Table 5: Wyckoff site: `8e`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,4]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[2,3]`
3	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	`[5,6]`
4	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[7,8]`

continued ...

Table 5

No.	position	mapping
5	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	[9,10]
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[11,12]
7	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	[13,16]
8	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[14,15]

Table 6: Wyckoff site: 8f, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{4}\right]$	[1,12]
2	$\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[2,9]
3	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[3,10]
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[4,11]
5	$\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$	[5,14]
6	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[6,15]
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[7,16]
8	$\left[-x,\ x,\ \frac{3}{4}\right]$	[8,13]

Table 7: Wyckoff site: 16g, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[2]
3	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[3]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$\left[y+\frac{1}{2},\ -x,\ -z\right]$	[5]
6	$\left[-y,\ x+\frac{1}{2},\ -z\right]$	[6]
7	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[7]
8	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[8]
9	$\left[\frac{1}{2}-y,\ x,\ z\right]$	[9]
10	$\left[y,\ \frac{1}{2}-x,\ z\right]$	[10]
11	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11]
12	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[12]
13	$[-x,\ -y,\ -z]$	[13]
14	$\left[\frac{1}{2}-x,\ y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[16]